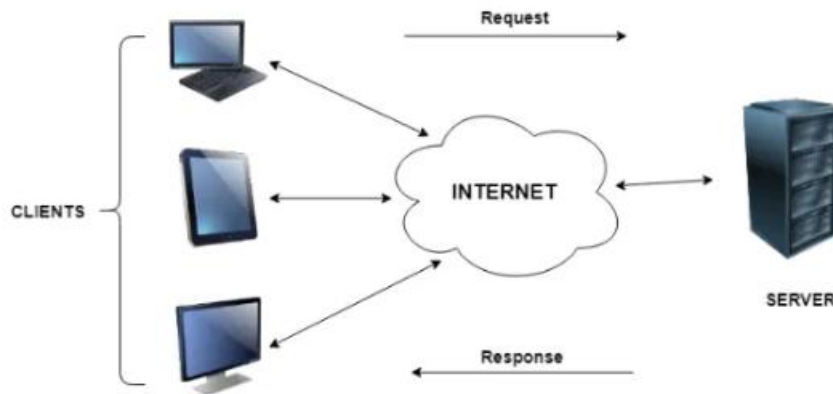


ASSIGNMENT – 2

Q) Draw and explain the client-server architecture and trace what happens when you open a website.



Client–Server architecture is a network model in which a **client** requests services or resources, and a **server** provides them.

- **Client:** End-user device or application (e.g., web browser like Chrome).
- **Server:** A remote computer that stores websites, applications, and data.
- Communication happens through **HTTP or HTTPS protocol** over the Internet.

The client sends a request, the server processes it, and then sends back a response. This request–response cycle is the core of web communication.

What Happens When You Open a Website?

Let's understand step by step what happens when you type a website URL in your browser:

1 Entering the URL

You type a domain name (e.g., www.example.com) in the browser.

2 DNS Resolution

The browser contacts the **Domain Name System (DNS)** to convert the domain name into an IP address, because servers are identified by IP addresses.

3 Establishing Connection

The browser establishes a TCP connection with the server using the obtained IP address. If it's HTTPS, a secure SSL/TLS handshake also happens.

4 Sending HTTP Request

The browser sends an HTTP request (like GET request) to the server asking for the webpage.

5 Server Processing

The server:

- Receives the request
- Processes it
- May interact with a database
- Generates the response (HTML, CSS, JavaScript files)

6 Server Sends Response

The server sends back the requested resources to the client.

7 Browser Rendering

The browser:

- Parses HTML
- Applies CSS styling
- Executes JavaScript
- Displays the final webpage on the screen